NAACL 2025

K-COMP: Retrieval-Augmented Medical Domain Question Answering With Knowledge-Injected Compressor

Jeonghun Cho **Gary Geunbae Lee**
Natural Language Processing Group @ POSTECH

POSTECH

Problem Statement

Challenges in closed-domain QA

- Documents retrieved for closed domains require domain expertise.

  Model exhibits an availability bias toward commonly known knowledge.
- Retrieved passages contain thousands of tokens and are unrelated to the question.

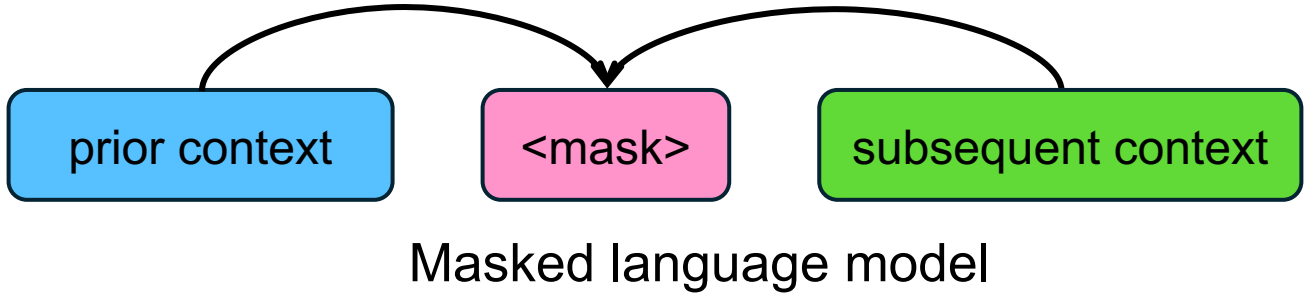
  This can cause the language model to distrust the passages.
- LLMs are sensitive to the order of retrieved passages and prompting methods.

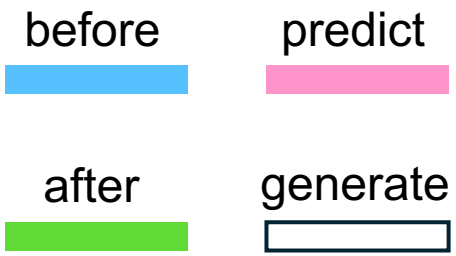
  LLMs can have difficulty finding the necessary information within lengthy prompts.

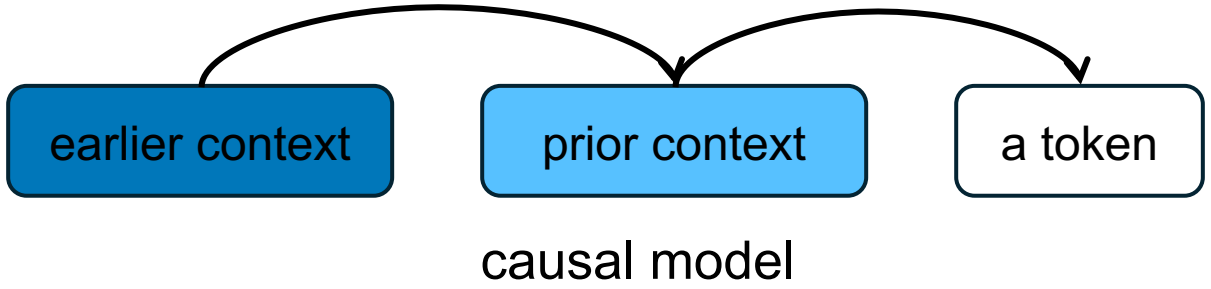
Proposed Method

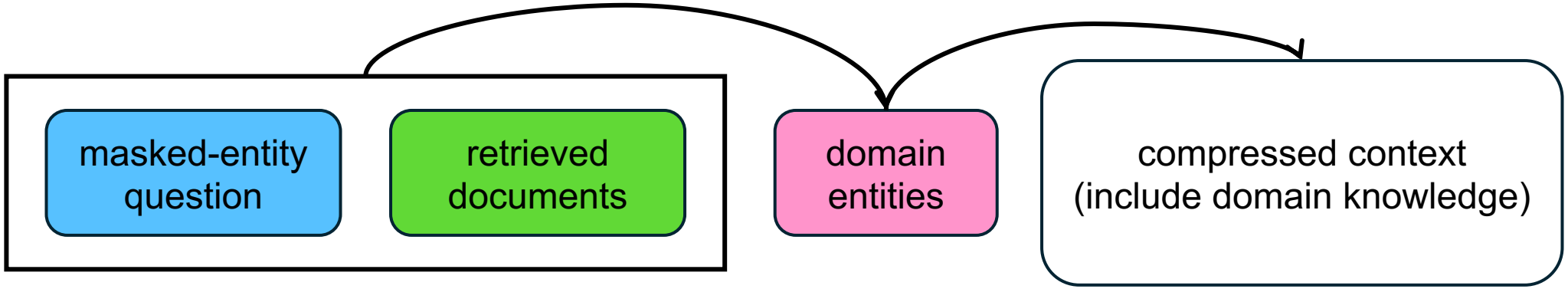
Causal Knowledge Injection

(inspired by Aghajanyan et al.)





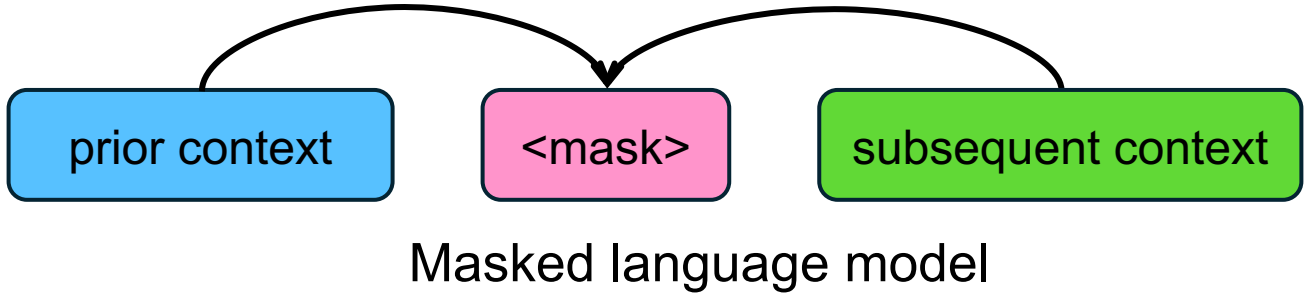


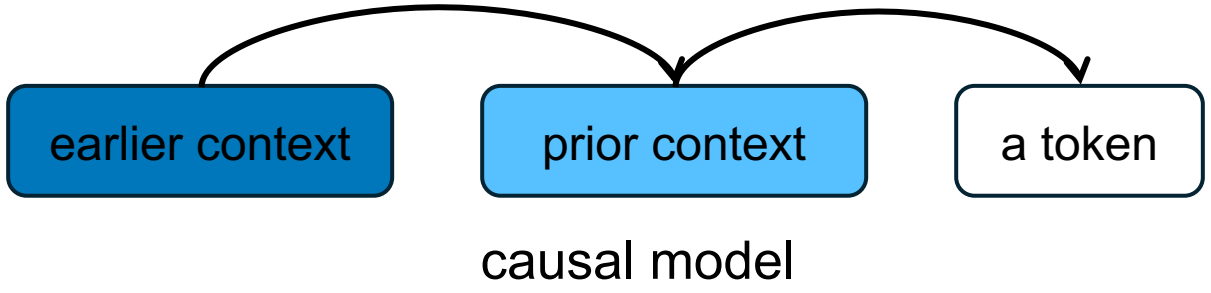


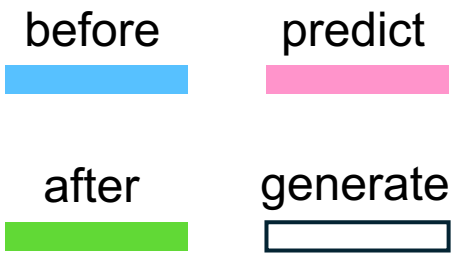
Proposed Method

Causal Knowledge Injection

(inspired by Aghajanyan et al.)







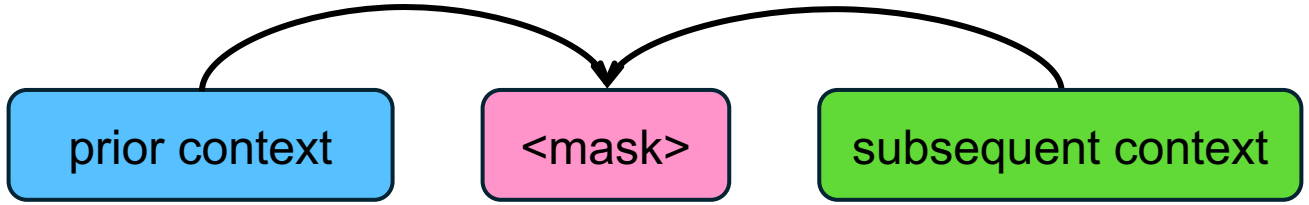
Unmasked question

retrieved documents

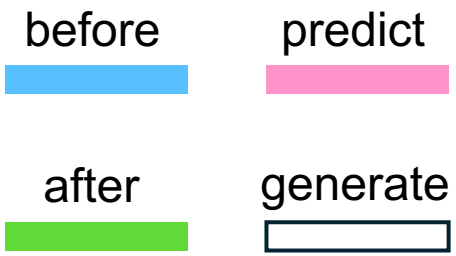
Proposed Method

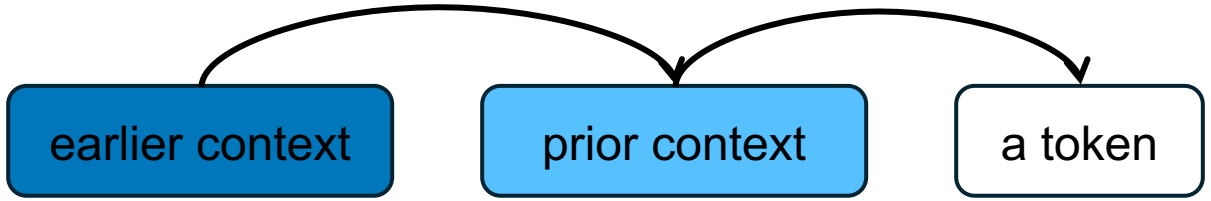
Causal Knowledge Injection

(inspired by Aghajanyan et al.)

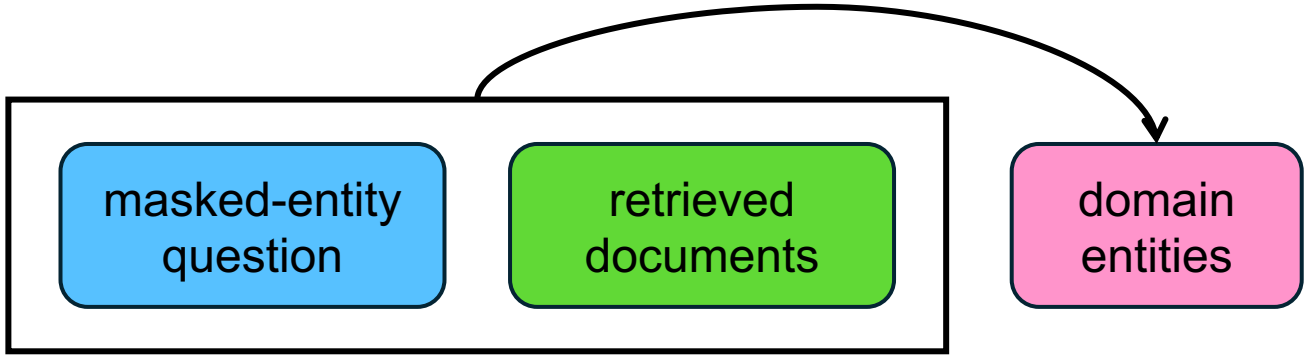


Masked language model





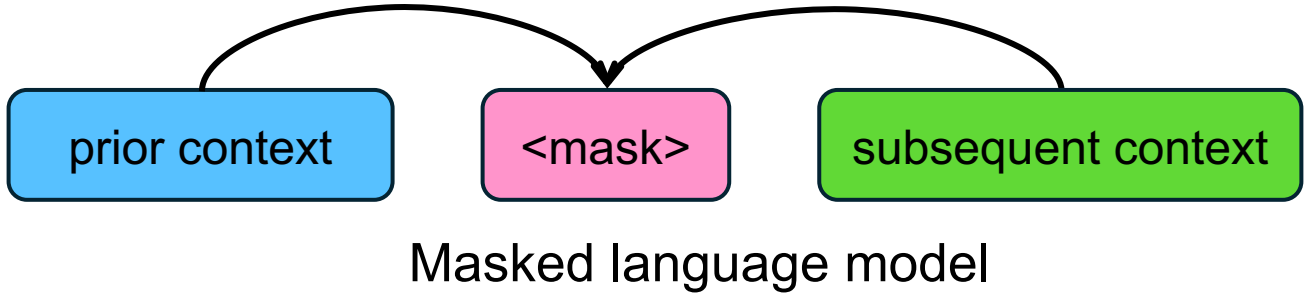
causal model

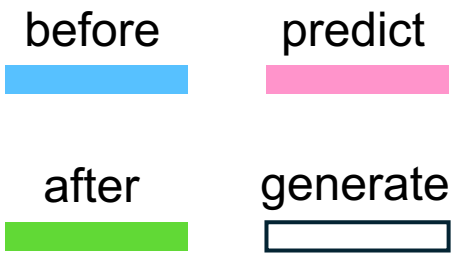


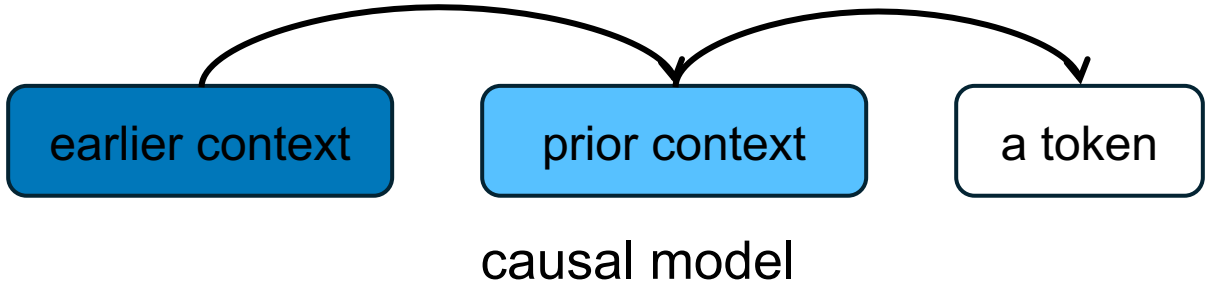
Proposed Method

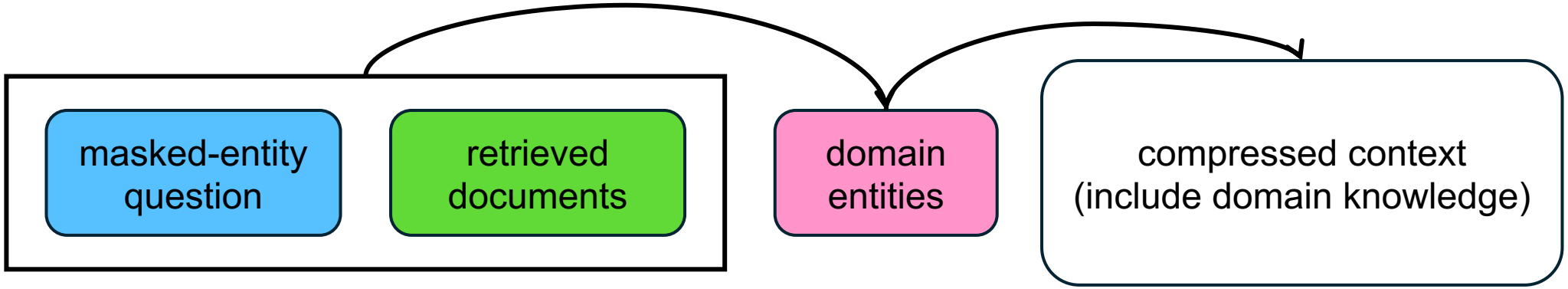
Causal Knowledge Injection

(inspired by Aghajanyan et al.)





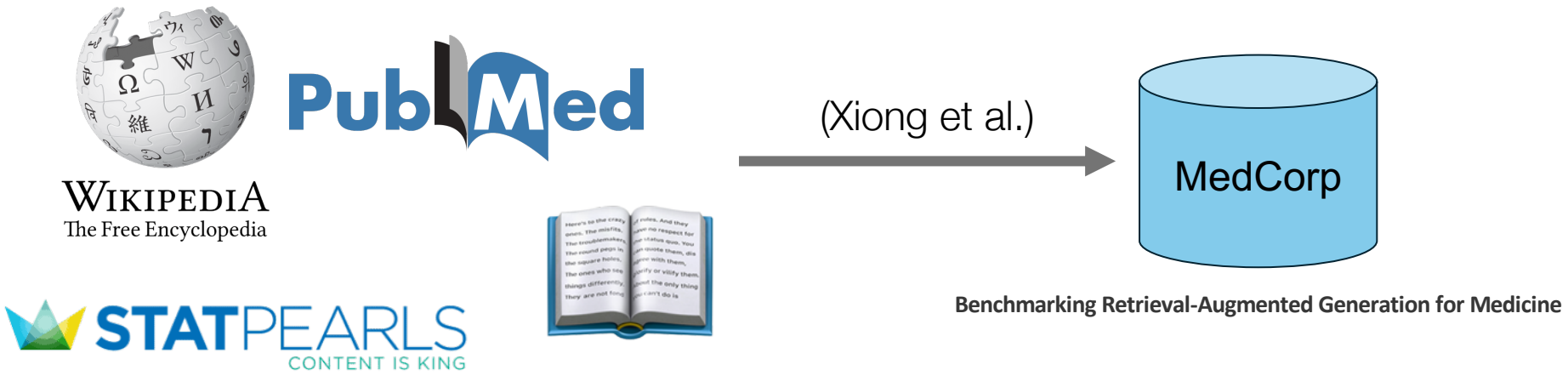




Prerequisites

Retrieval Corpus for retrieval-augmentation

- To ensure of both general and domain knowledge, we adopt the MedCorp (Xiong et al.).



Prerequisites

Retriever for relevant context selection

- Retrieval Corpus for retrieval-augmentation
  To ensure of both general and domain knowledge, we adopt the MedCorp (Xiong et al.).

- We employ an embedding-based k-NN search model (i.e., encoder models).

Retriever Model

Any retriever is fine.

(We use Nomic Embed (Nussbaum et al.))

Prerequisites

Data processing for fine-tuning

- Retrieval Corpus for retrieval-augmentation
  To ensure of both general and domain knowledge, we adopt the MedCorp (Xiong et al.).

- Retriever for relevant context selection
  We employ an embedding-based k-NN search model (i.e., encoder models).

- Ground-truth summaries are synthesized using gpt-4o-mini, and entities are recognized with spaCy-based tool.

$$q = \text{What are the symptoms of down syndrome ?}$$

$$q_m = \text{What are the <ent> of <ent>?}$$

(Neumann et al.)

Dataset Preview

- Masked-entity question & Entities description

$q = \text{What are the symptoms of down syndrome ?}$

$q_m = \text{What are the <ent> of <ent>?}$

$d_1 = \text{symptom: \{description\}<eod>}$

$d_2 = \text{Down syndrome: \{description\}<eod>}$

- Passages & Domain Knowledge

$p_i = \text{retrieved passage}$

$e_i = \text{entity included in the question (e.g., symptom)}$

$s = \text{ground}-\text{truth summary}$

$\mathbb{E} = \{e_1 \dots e_n\} \qquad \mathbb{D} = \{d_1 \dots d_n\} \qquad \mathbb{P} = \{p_1 \dots p_5\}$

Training method

Supervised Fine-Tuning

$$P_\theta(\mathbb{E}, \mathbb{D}, s \mid q_m \oplus \mathbb{P}) = P_\theta(\mathbb{E}, \mathbb{D} \mid q_m \oplus \mathbb{P}) \times P_\theta(s \mid \mathbb{E}, \mathbb{D}, q_m \oplus \mathbb{P})$$

$$\therefore L(\theta) = -E(\log P_\theta(\mathbb{E}, \mathbb{D}, s \mid q_m \oplus \mathbb{P}))$$

Overall Performance

	General-purpose LLMs								Medical-purpose LLMs			
	Llama-3-8B		Llama-3-70B		Mixtral-8x7B		GPT-4o [7]		MedAlpaca-13B		Meditron-70B	
	BertScore	UniEval	BertScore	UniEval	BertScore	UniEval	BertScore	UniEval	BertScore	UniEval	BertScore	UniEval
MedQuAD												
Without compressor												
Top-1 document	75.65	53.44	79.44	57.47	58.68	44.30	84.45	60.24	75.05	36.89	76.38	52.74
Top-5 documents	76.20	52.45	79.95	55.25	64.17	47.47	83.08	**64.98**	17.57	10.82	73.14	51.88
With compressor												
RECOMP	73.27	55.74	79.75	**63.57**	71.58	58.78	85.20	60.92	82.52	39.15	76.41	56.14
LLMLingua	76.45	54.65	78.78	57.00	59.79	45.34	84.46	55.04	83.10	44.17	74.74	54.18
FineTune	74.27	54.89	79.97	59.67	71.44	56.26	83.47	62.40	82.25	39.40	76.14	55.97
K-COMP	**83.79**	**64.05**	**82.06**	62.91	**80.64**	**63.07**	**85.49**	63.16	**85.75**	**49.39**	**78.88**	**58.95**
MASH-QA												
Without compressor												
Top-1 document	77.87	54.08	82.65	56.48	72.63	51.88	83.39	60.66	76.95	43.41	80.45	55.31
Top-5 documents	77.59	52.94	82.07	57.20	73.09	53.18	83.12	**64.81**	31.91	20.07	79.31	55.85
With compressor												
RECOMP	80.20	58.80	82.27	60.50	**82.12**	62.35	83.82	62.22	78.51	34.83	81.54	60.09
LLMLingua	79.74	54.37	83.25	56.83	76.71	58.32	83.37	56.73	80.41	48.45	80.78	58.55
FineTune	77.30	55.21	82.90	60.66	78.74	60.62	83.88	62.82	79.96	44.02	81.70	60.06
K-COMP	**83.45**	**61.45**	**83.64**	**61.40**	81.49	**62.91**	**83.92**	64.06	**81.95**	**51.76**	**82.67**	**61.85**
BioASQ												
Without compressor												
Top-1 document	82.53	56.03	81.40	59.58	82.23	62.07	87.13	62.47	76.86	40.06	83.43	57.55
Top-5 documents	81.44	61.33	80.91	61.09	83.54	65.53	86.81	**66.15**	20.93	12.41	83.63	59.52
With compressor												
RECOMP	83.89	59.80	85.45	59.65	**86.85**	62.82	87.87	60.62	78.07	30.68	84.77	57.58
LLMLingua	81.70	53.62	86.53	61.53	82.54	62.24	86.85	51.11	81.19	42.97	81.51	55.26
FineTune	82.26	60.89	86.17	62.86	85.48	64.01	**87.97**	63.99	78.67	41.49	85.05	61.20
K-COMP	**86.61**	**62.28**	**87.03**	**62.89**	86.52	**66.31**	87.77	64.38	**84.88**	**45.47**	**86.70**	**62.90**

Table 1: Main results. We report automatic evaluations for retrieval-augmented QA with and without compressors.

Entity's description is effective ?

	Llama-3-8B		Llama-3-70B		MedAlpaca-13B		Meditron-70B	
	BertScore	UniEval	BertScore	UniEval	BertScore	UniEval	BertScore	UniEval
				MedQuAD				
K-COMP	**83.79**	**64.05**	**82.06**	**62.91**	**85.75**	**49.39**	78.88	58.95
$-Prior$	73.93	54.47	80.06	60.03	84.84	48.23	**79.80**	**59.00**
FineTune	74.27	54.89	79.97	59.67	82.25	39.40	76.14	55.97
				MASH-QA				
K-COMP	**83.45**	**61.45**	**83.64**	**61.40**	81.95	51.76	**82.67**	**61.85**
$-Prior$	78.63	57.80	83.16	60.88	**82.18**	**52.02**	81.93	60.42
FineTune	77.30	55.21	82.90	60.66	79.96	44.02	81.70	60.06
				BioASQ				
K-COMP	**86.61**	**62.28**	**87.03**	**62.89**	**84.88**	45.47	**86.70**	**62.90**
$-Prior$	80.75	55.06	86.22	62.40	84.53	**46.67**	85.23	59.73
FineTune	82.26	60.89	86.17	62.86	78.67	41.49	85.05	61.20

Table 2: Ablation studies. $-Prior$ denotes the scenario where K-COMP does not provide prior knowledge to the reader LLMs.

Reranking Preference

We employ a state-of-the-art reranker to measure the relevance between the context and the question.

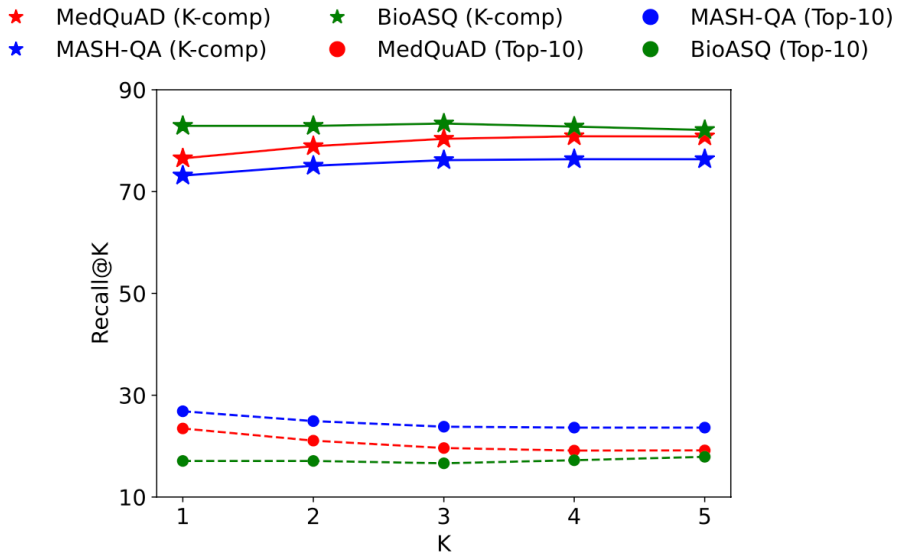


Figure 2: Percentage of Recall@K according to the variation of K for the retrieved passages and our compressed contexts, where Top-10 denotes the ten retrieved passages with the highest similarity scores.

Inference Speed

Inference speed is dependent on the implementation and size of the reader model.
This phenomenon amplifies the speed advantage of K-COMP.

Settings	Top-1	Top-5	K-COMP
Input tokens	195	978	198
Inference time	9,798s	20,193s	9,777s
Compression time	-	-	497s
Total time	9,798s	20,193s	10,274s

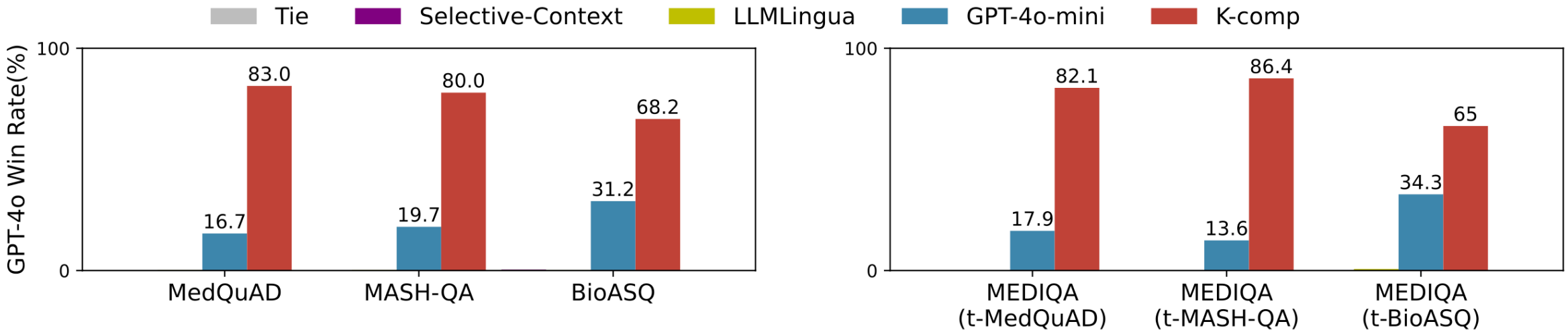
Table 3: Inference speed of Llama-3-70B on MASH-QA.

Case Study

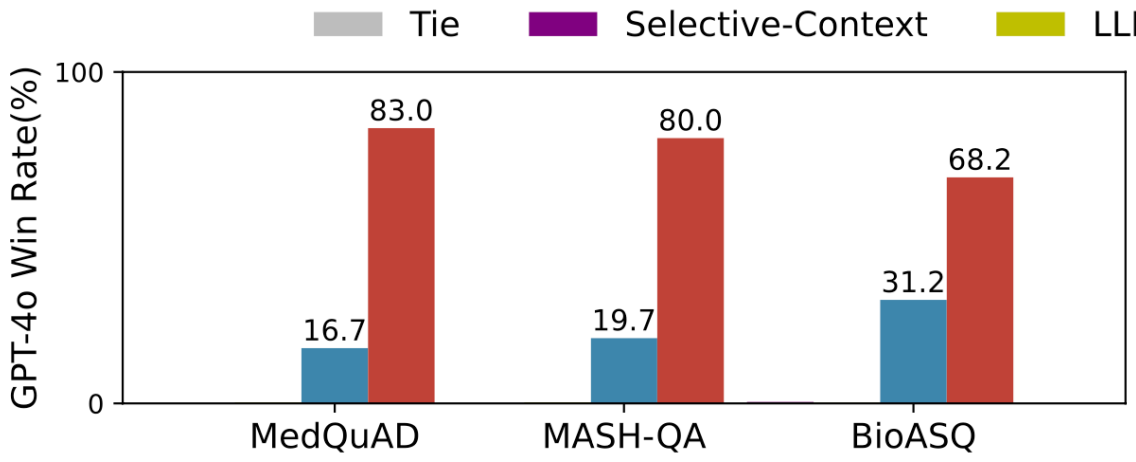
	Question: How do **x-rays** diagnose **rheumatoid arthritis** (RA)?	
	Augmented Prompt	Answer
FineTune	### Passage Rheumatoid arthritis (RA) is a chronic systemic disease characterized by inflammation of the synovial tissue, affecting 0.5%-1.0% of the global population, with ...	X-rays are used to diagnose RA by detecting joint damage and erosion.
K-COMP	### Entity **x-ray**:{{description}} **rheumatoid arthritis**: {{description}} ### Passage **X-rays** are the most common method for assessing the degree of joint destruction in **rheumatoid arthritis (RA)**, revealing osteopenia and joint space narrowing as the disease ...	X-rays diagnose rheumatoid arthritis (RA) by revealing changes in the axial skeleton and joint space. As the disease progresses, early RA may not show significant changes on X-rays, but it can indicate osteopenia and joint space narrowing. Advances in imaging techniques like MRI and ultrasound enhance the ability to detect soft tissue changes and joint damage, which is crucial for accurate diagnosis ...

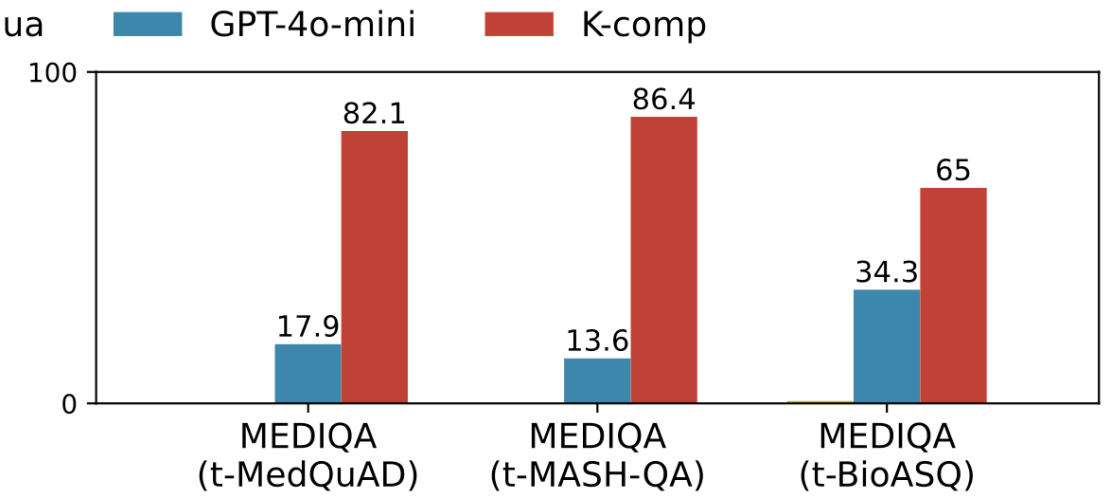
Table 4: Case study. We provide the passages used to augment the reader's prompt and the answers. **Red texts** highlight the medical jargon within the question. The complete prompt can be found in Table 14.

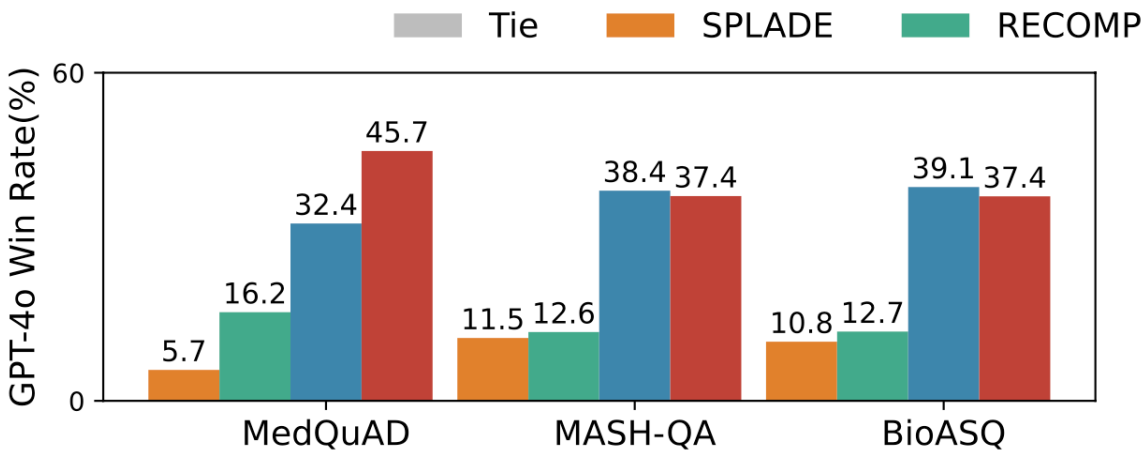
GPT-4o Evaluation

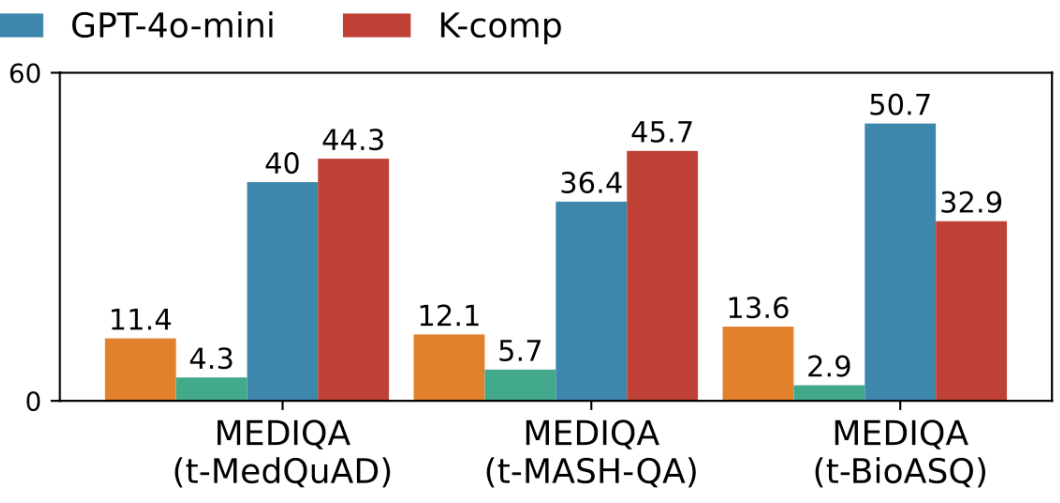


GPT-4o Evaluation









GPT-4o Evaluation

G-Eval (Liu et al.)

	Llama-8B	Llama-70B	Mixtral	GPT-4o	MedAlpaca	Meditron
			MedQuAD			
Without compressor						
Top-1 doc	2.96	3.11	2.63	4.27	1.65	3.14
Top-5 docs	3.22	3.36	3.07	**4.73**	1.85	3.53
With compressor						
RECOMP	3.29	**3.80**	3.51	3.92	1.48	3.68
LLMLingua	2.41	2.87	2.53	3.68	1.61	3.21
FineTune	3.28	3.66	3.27	4.15	1.44	3.68
K-COMP	**3.55**	3.66	**3.73**	4.48	**1.98**	**3.85**
			MASH-QA			
Without compressor						
Top-1 doc	3.07	3.65	3.24	4.63	1.64	3.55
Top-5 docs	3.33	3.58	3.46	**4.74**	1.79	3.43
With compressor						
RECOMP	3.68	4.14	**4.49**	4.15	1.34	4.01
LLMLingua	2.97	3.18	3.77	3.99	1.88	3.47
FineTune	3.36	3.79	3.78	3.93	1.74	3.51
K-COMP	**4.23**	**4.62**	4.06	4.52	**1.99**	**4.53**
			BioASQ			
Without compressor						
Top-1 doc	3.44	3.75	3.64	4.33	2.09	3.76
Top-5 docs	3.58	3.66	3.68	**4.53**	1.32	3.64
With compressor						
RECOMP	3.64	3.92	**4.10**	3.58	1.18	4.00
LLMLingua	2.54	3.63	3.54	3.37	1.87	3.42
FineTune	3.63	4.05	3.59	4.01	1.48	3.88
K-COMP	**4.10**	**4.14**	3.81	4.20	**2.31**	**4.38**

Table 5: Results evaluated using the G-Eval-4 metric (Liu et al., 2023b). The prompt used is presented in Table 13.

Summary

We introduce K-COMP, a knowledge-injected prompt compressor.

- Question-focused summary generation.
- Improved QA performance in closed-domain settings.
- Strong generalization to unseen, low-resource environments.

E-mali: jeonghuncho (at) postech.ac.kr

Homepage: https://github.com/jeonghun3572